Bhavik Barot

6062 Avenue McDonald, Montreal, QC H3X 2X1

438-979-4921 — bhaviksbarot@gmail.com — Linkedin — GitHub

Summary

Mechanical/ Aerospace Engineer (M.Eng., Concordia University) with experience in stress, fatigue, thermal, and seismic analysis using ANSYS APDL/Workbench, MATLAB, and MathCAD. Professional background includes pressure equipment design and aerospace manufacturing at Safran Aerospace and Pratt & Whitney Canada, with strengths in QA documentation, NCR resolution, and compliance reviews. Eligible for PEO EIT, motivated to advance in nuclear pressure boundary design and analysis.

Skills

- **Programming & Data Analysis**: Python, MATLAB, Cantera (Combustion Modeling), Power BI, VBA, MySQL
- **Simulation & CAD Tools**: AutoCAD, Solidworks, CATIA V5 (3-D Modeling), Siemens NX, ANSYS (Fluent, Structural, Classic), StarCCM+, Nastran, Patran, ANSYS APDL, Simulink
- **Collaboration & Productivity Tools**: Microsoft Office, Power BI, SharePoint, SAP, Enovia, ESYNC, Jira
- **Codes & Standards:**:Familiar with ASME BPVC Sec VIII, exposure to ASME Sec III, B31.1, CSA B51/N285.0.
- **Languages**: English, French (TEFAQ-B2 Level)

Experience

Safran Aerospace - Kirkland, Quebec **March - 2025**
Method Designer

- Leading a project to automate manufacturing work instruction (MWI) development by detecting iterative design changes using Python-based scripts, enhancing traceability and reducing manual rework.
- Collaborated with design, engineering, and production teams to interpret Condition of Supply (CSD), BOMs, and 3D/2D models in CATIA V5.
- Validated design updates, resolved geometrical deviations and NCRs, and maintained QA documentation in compliance with aerospace standards.
- Supported engineering change packages by preparing technical notes and ensuring controlled documentation for regulatory compliance.

Pratt & Whitney Canada - Longueuil, Quebec **May 2024 – Sept 2024**
Operations Program Analyst (Full-Time Internship)

- Supported engine repair and overhaul activities by ensuring materials met high-performance standards. Collaborated with cross-functional teams to address material-related challenges and ensured alignment with design specifications.
- Developed a Project Management Dashboard POWER BI and Power Automate, automating processes and enabling real-time tracking to support operational excellence through Continuous Improvement.
- Participated in Integrated Product Team (IPT) meetings with engineering, quality, and supply chain to review NCRs, source changes, and documentation for regulatory adherence.

Vitthalesh Engineering -Gujarat, India **Aug 2021 – Aug 2022**
Graduate Engineer – Mechanical Design & Simulation

- Designed refinery pressure boundary components (vessels, piping supports, skid assemblies) in SolidWorks supplied to Larsen & Toubro (L&T) for oil & gas projects and prepared GA drawings, ISO drawings, and BOMs for manufacturing.
- Performed stress, fatigue, and thermal analysis in ANSYS APDL/Workbench on welded joints, brackets, and nozzle connections; validated results against ASME BPVC Sec. VIII allowables.
- Performed stress, fatigue, and thermal analysis in ANSYS APDL/Workbench on welded joints, brackets, and nozzle connections; validated results against ASME BPVC Sec. VIII allowables.
- Collaborated with QA and fabrication teams to resolve NCRs and deviations, maintaining compliance with ASME and client specifications.
- Prepared technical documents and inspection packages for refinery pressure equipment in compliance with ASME Section VIII.

Larsen & Toubro Defence IC -Gujarat, India **Jan 2021 – Aug 2021**
Mechanical Design Intern

- Designed 3D models and 2D drawings in CATIA V5 as per GD&T and supportability standards, collaborated with production to ensure manufacturability, compliance, and efficient logistics through transportation considerations.
- Applied Model-Based Systems Engineering (MBSE) to define and communicate system architectures for accessory component integration, improving coordination across design, analysis, and manufacturing in the product lifecycle.

Education

Concordia University **Sept 2022 – Sept 2024**
M.Eng Aerospace Engineering
Major : Combustion, Thermofluid Science *Major : Dynamics of Combustion (IUT-McGill University), Thermofluid Science*

Gujarat Technological University **July 2017 – June 2021**
B.E. Mechanical Engineering
Major : Mechanical Engineering

Projects

Design Optimization of High-Pressure Turbine and Compressor Blade **Pratt & Whitney Canada**

- **Objective**: Optimize aerodynamic and structural performance of high-pressure turbine and centrifugal compressor blades under stress, fatigue, and durability constraints.
- **Solution**: Used Python-based numerical optimization to explore design space and refine blade geometry. Modeled 3D blade profiles in CATIA V5 and conducted CFD and structural analysis in ANSYS Fluent to assess life cycle and off-design behavior.
- **Result**: Delivered an optimized blade design that met stress, material durability, and aerodynamic efficiency targets, enhancing performance and extending component life.

Design of Low emission Fuel Injection System for Gas Turbine **Concordia University**

- **Objective**: Designed a low-emission hydrogen-fueled injector for gas turbines, minimizing NOx emissions while addressing operational needs and optimizing combustor performance.
- **Solution**: Incorporated micromixing and multipoint injection methods using ANSYS Fluent for CFD simulations and performance analysis.Performed extensive testing to validate the design criteria.
- **Result**: Resulted in a validated low-NOx injector supporting Siemens' goals in clean propulsion technology.

CFD Simulation and Performance Analysis of VKI High Pressure Turbine Blade **Concordia University**

- **Objective**: Conduct 3D aerodynamic simulations of a VKI turbine blade cascade to analyze pressure distribution, flow-behavior, and performance sensitivity using ANSYS CFX and StarCCM+.
- **Solution**: Developed a parametric cascade model with structured mesh and applied realistic boundary conditions. Simulated compressible flow across various Mach numbers and analyzed key parameters like pressure coefficient, velocity,and isentropic Mach using post-processing tools.
- **Result**: Validated CFD models and extracted insights to support blade shape optimization and performance evaluation under off-design conditions.

Implementation of RQL (Rich Burn Quick Lean) Combustor in Cantera Modeling **McGill University**

- **Objective**: To analyze the feasibility of hydrogen as an alternative fuel by blending it with jet fuel and visualizing the combustion dynamics of the blended fuel with varying proportions.
- **Solution**: Modeled hydrogen-blended jet fuel combustion using Cantera and Python, quantifying risks and demonstrating its feasibility to minimize harmful emissions and enhancing operational efficiency.
- **Result**: The simulation demonstrated that the mixture of hydrogen and jet fuel with optimum mixing ratio has the potential to significantly reduce NOx emissions, making hydrogen a viable alternative fuel option for cleaner combustion processes.

Certifications & Achievements

- **Certification Training of Microsoft POWER BI for Project Management** (2023)
- **Selected for the prestigious Princeton University-CEFRC (Combustion Energy Frontier Research Center) Summer School on Combustion (2023)**
- **Aerodynamic CFD Simulation Training Course by ANSYS Fluent (In Progress)**
- **Finite Element Analysis using ANSYS APDL (In Progress)**
- **Short Term Training Program (STTP) On Pedagogy and Research Methodology Sponsored by GUJCOST (2021)**
- **CATIA V5 Complete Training – Udemy**
- **Active member of ASME under student membership program**
- **How to write & publish Scientific paper (Project-Centered Course)(2020)**
- **MATLAB Certification from UDEMY (2022)**

Research Paper

Conducted research on micromix combustion as a low-NOx strategy for gas turbines by analyzing the aerodynamic behavior of hydrogen flamelets. Focused on flame structure, stabilization, and jet-in-crossflow interactions to assess flame anchoring, temperature uniformity, and NOx reduction. Compared micromix performance with DLN (Dry Low NOx) and RQL (Rich Burn Quick Lean) technologies from an aero-thermal standpoint, highlighting benefits in stability, mixing efficiency, and pressure loss control.